ADITYA MALVIYA

SENIOR DATA ENGINEER | AZURE DATABRICKS | PYSPARK | DELTA LAKE

Mumbai, India | Phone: +91 8555981411 | Email: adityamalviya04692@gmail.com

linkedin.com/in/adityamalviya04692 | github.com/adityamalviya04692 | adityamalviya04692.github.io/aditya-portfolio

PROFESSIONAL SUMMARY

Senior Data Engineer with nearly 5 years at Tata Consultancy Services (TCS) building production data platforms for Rolls-Royce and Avery Dennison. Designs batch and streaming ETL/ELT pipelines (~27 GB/day) in Python, SQL and PySpark on Azure Databricks, Azure Data Factory, ADLS Gen2, Delta Lake and Unity Catalog using Medallion Architecture and dimensional data modeling. Cut Spark runtime by 73%, reduced Azure costs by about 40%, and leads a 6-member offshore team as Scrum Master.

TECHNICAL SKILLS

Programming & Processing: Python, SQL, T-SQL, PySpark, Apache Spark, Spark SQL, Spark Structured Streaming, Spark Performance Tuning
Azure & Databricks: Azure Databricks, Azure Data Factory (ADF), Azure Data Lake Storage Gen2 (ADLS Gen2), Azure Event Hubs, Apache Kafka, Delta Lake, Unity Catalog, Databricks Workflows, Auto Loader, Azure Key Vault, Azure DevOps
Google Cloud & AI: Google Cloud Platform (GCP), Databricks on Google Cloud, Vertex AI, Google Cloud Functions, Vector Databases
Data Architecture & Modeling: ETL, ELT, Data Pipelines, Lakehouse Architecture, Medallion Architecture, Data Warehousing, Dimensional Data Modeling, Star Schema, SCD Type 2, Incremental Loads
Data Governance & Quality: Data Governance, Data Quality Frameworks, Data Lineage, Role-Based Access Control (RBAC)
Delivery & Leadership: Scrum Master, Team Leadership, Agile, Code Reviews, CI/CD, Git, YAML, Docker, Pytest, SonarQube, GitHub Copilot, Claude, ChatGPT
Databases, APIs & BI: Microsoft SQL Server, MySQL, PostgreSQL, Redis, FastAPI, REST APIs, Azure API Management, Power BI
Working knowledge: Apache Airflow, Azure Synapse Analytics, Delta Live Tables

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) | Mumbai, India **Dec 2021 - Present**
Senior Data Engineer **Mar 2026 - Present**
Data Engineer **Dec 2021 - Feb 2026**

Client: Rolls-Royce Digital (global aerospace) ***Feb 2022 - Present***

Project: Engine History Analysis - Azure Lakehouse Platform **2024 - Present**

- Design and own production **Azure Data Factory (ADF)** and **Azure Databricks** ETL/ELT pipelines in **PySpark** and Spark SQL on a **Medallion Architecture** (Bronze, Silver, Gold) over Delta Lake and ADLS Gen2, processing **~27 GB/day of streaming** and **5-7 GB/day of batch** engine data into star-schema models for reporting.
- Built real-time ingestion with **Azure Event Hubs**, **Apache Kafka** and **Spark Structured Streaming** into Delta Lake, alongside incremental batch loads using Delta MERGE, **SCD Type 2** and Auto Loader, orchestrated with Databricks Workflows.
- Cut a critical Spark job's runtime by **73% (52 to 14 minutes)** by diagnosing data skew in high-volume joins and applying key salting and Adaptive Query Execution (AQE), eliminating recurring executor out-of-memory failures.
- Reduced Azure costs by **about 40%** through Databricks cluster right-sizing, Delta Lake OPTIMIZE, Z-ORDER, partitioning and parameterized ADF pipelines.
- Implemented layer-wise data quality checks and Unity Catalog governance (RBAC, data lineage); secured credentials with Azure Key Vault secret scopes and service principals.
- Since 2025, lead a **6-member offshore team as Scrum Master**: run daily stand-ups and sprint planning, maintain Azure DevOps boards, mentor juniors and approve their PySpark and SQL pull requests, and own delivery commitments to the client.
- Use **AI-assisted development** (GitHub Copilot, Claude and ChatGPT in VS Code) to automate repetitive coding and test writing and to raise code quality across the team.

Project: Assumptions Management - Data Services and APIs **Feb 2022 - 2024**

- Built production FastAPI (Python) data services with RBAC, Azure API Management and Docker over SQL Server and Blob Storage; improved performance by **about 30%** with Redis caching and SQL/T-SQL optimization.

Client: Avery Dennison - Smartrac RFID division (US Fortune 500) ***Dec 2021 - Jan 2022***

Project: Smartrac R&D Data and AI Platform - Google Cloud and Databricks

- Built **Databricks** data engineering pipelines on **Google Cloud Platform (GCP)** to ingest, cleanse and prepare Smartrac RFID R&D datasets used to train machine learning models for an AI-powered research tool, now live in production.
- Developed backend services and AI components with **Vertex AI** (Google pre-trained models), Cloud Functions and a vector database for semantic search.

PROJECTS

Independent Research: Spark Skewed-Join Benchmark (AQE vs Key Salting) **2026**

- Built a reproducible PySpark benchmark (6 join strategies, Spark event-log parser, 55 Pytest tests) showing AQE's skew-join rule never fired when advisoryPartitionSizeInBytes exceeded the hot partition, while selective salting cut the critical path **1.9x**.

CERTIFICATIONS

Microsoft Certified: Fabric Data Engineer Associate (DP-700) | Microsoft Certified: Azure Data Fundamentals (DP-900)

EDUCATION

Bachelor of Technology (B.Tech), Mechanical Engineering | Bhilai Institute of Technology, Durg | 71% **2021**